

WI-FI SENSING: Smart Tracking With No Wearables, No Cameras, And No Privacy Worries!

What if your Wi-Fi could sense motion, track location, and detect actions—without cameras or wearables? CSI makes it possible, quietly and smartly.



DR SUDHIR KUMAR is Associate Professor at the Department of Electrical Engineering, IIT, Patna



AVNISH ARYAN is pursuing BTech and undergraduate research in Electrical and Electronics Engineering in the Department of Electrical Engineering, IIT, Patna

The Internet of Things (IoT) is reshaping how we interact with our environment—from adjusting lights to monitoring patient health. Meaningful interaction requires awareness of both human activity and location. While activity recognition detects what someone is doing, localisation identifies where it is happening. Tracking integrates the two.

Indoor cameras can serve this purpose but raise privacy concerns and incur installation costs. A more seamless alternative leverages existing Wi-Fi signals. Unlike GPS, which struggles indoors due

to obstructions, Wi-Fi-based channel state information (CSI) offers granular insights into signal fluctuations caused by physical movement. These variations reveal motion, specific actions, and precise locations—without relying on cameras or wearables.

As Wi-Fi signals traverse a space, they encounter obstacles like walls, furniture, and human bodies, altering amplitude and phase. CSI captures these shifts, providing fine-grained data about the wireless channel. Analysing these changes enables passive detection of movement, recogni-

tion of specific activities, and indoor localisation—without the need for additional hardware.

Application

CSI supports passive monitoring that is especially effective in healthcare and smart living environments. In elderly care, it detects falls or irregular movements without wearables, preserving dignity while enabling continuous oversight.

In smart homes, CSI can identify

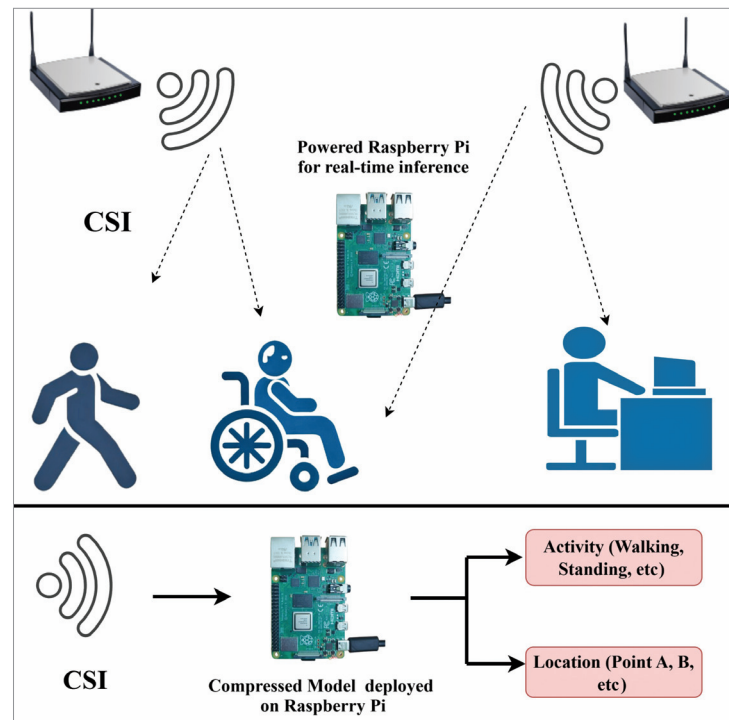


Fig. 1: Overview of CSI-based joint activity recognition and localisation on edge